JAIKRISHNA SEKARAN

+91 94896 63387 Salem, Tamil Nadu DOB: 24 March 2006
jaikrishnasekaran@gmail.com ◇ LinkedIn

PROFILE SUMMARY

Fourth-year B.Tech Computer Science student with a strong interest in AI/ML and hands-on experience building Python-based projects. Passionate about applying machine learning and data-driven techniques to solve real-world problems, with a solid foundation in algorithms, programming, and experimentation.

EDUCATION

Bachelor of Technology in Computer Science Engineering Aug 2023 - June 2027 (expected)
Amrita Vishwa Vidyapeetham *Coimbatore*
CGPA: **8.28**

Grade 12 June 2022 - Mar 2023
GlazeBrooke Public School *Salem*
Percentage: **93.8%**

Grade 10 Completed 2021
GlazeBrooke Public School *Salem*
Percentage: **95.8%**

Standardized Testing
JEE Mains: **91.7 Percentile**

RESEARCH INTERNSHIPS

SONY SSUP AgroESP Research Project – Model Developer Aug 2024 – Aug 2025
Worked on developing and integrating machine learning models for estimating food drying processes in a polyhouse environment. Contributed to data analysis, image-based modeling, and optimization under embedded system constraints using the Sony Spresense platform.

SyberGate Technologies Pvt. Ltd. – Software Developer Sept 2025 – Feb 2026
Developing simulation models for the chlor-alkali process in Python using Gmsh and FEniCSx, focusing on mathematical modeling and scalable system design. Engineered non-uniform layered meshes (Anode, Membrane, Cathode) to accurately simulate ion movement, successfully resolving concentration drops and mitigating sink-like visual signatures. Leveraging multiprocessing for efficient computation and building interactive data visualization dashboards with Plotly, Dash, and Streamlit for monitoring and analyzing simulation results.

Landcheck - AI Engineer Mar 2026 - Jun 2026
Developed a legal-document RAG system from scratch for TNRERA complaint records, implementing automated web scraping, OCR extraction and noise cleaning, metadata-aware chunking, FAISS-based retrieval, Gemini-powered semantic search, and Streamlit deployment. Processed over 1,300 lines of complex OCR-extracted legal data, carefully preserving critical identifiers like dates and names. Engineered dynamic content generation, interval-tree scheduling, and JSON-based content management, utilizing relative file paths for unique identification to prevent data collisions across workflows.

PROJECTS

Minimal Spectral Feature Discovery for AI-based Land Cover Classification
Engineered a scalable framework to exhaustively evaluate Sentinel-2 spectral band combinations for land cover classification, identifying minimal and non-conventional feature sets through systematic experimentation. Optimized data pipelines to automate hundreds of reproducible training runs with efficient benchmarking.

A Computer Vision-Based Human Fall Detection System *[GitHub]*
Built a five-stage multimodal pipeline combining 3D-CNN spatio-temporal localization, YOLOv8m-Pose, MiDaS depth estimation, and BiLSTM-Attention temporal modeling for fall detection. Developed a physics-guided decision engine that reduced false positives by 59% (84.3% accuracy on cross-dataset evaluation). Further expanded the system by training and optimizing ensemble models (Random Forest, SVM, Voting Ensemble) on UP-Fall and CAUCAFall datasets to achieve stage-specific accuracies between 87% and 89%.

AI-Based Crop Disease Detection and DSS Remedy Recommendation System *[GitHub]*
Worked as part of a team to design and implement a deep learning pipeline for crop and crop disease classification. Contributed to data preprocessing, model training, evaluation, and remedy recommendation logic.

BlueprintGPT *[GitHub]*
Hybrid floor-plan generator combining a transformer-based LayoutTransformer, deterministic planners, and ontology-backed rules to generate compliant architectural layouts. Uses learned bounding-box priors for polygon packing (Shapely), followed by deterministic repair, door/corridor placement, ranking, and guarded LLM explanations.

KnowYourFunctions
Created a graph database utilizing the official NumPy manual as ground truth to accurately fetch and display relevant functions for user queries. Designed to prevent AI hallucination and establish seamless connections across multiple libraries and programming languages.

TrackLearn
Implemented epsilon weight decay reinforcement learning to enable an autonomous vehicle agent to learn and navigate a track, fully simulating the physics and results within the Webots environment.

SecureEdu Comm Platform
Developed a cryptographically secure teacher-student communication platform demonstrating the practical application of RSA-256, salted hashing, robust Encoding/Decoding, and Authentication/Authorization through Digital Signatures.

C-Signal Processing
Programmed various fundamental signal processing techniques entirely from scratch in C, implementing visualization techniques to graph and analyze signal transformations.

Driver Fatigue Edge Detection
Engineered a real-time computer vision sleep detection module and deployed it on an STM32 edge device to infer driver fatigue through image analysis and instantly alert the driver.

Interval-Tree Event Manager
Developed an efficient Event Management System leveraging Interval Tree data structures to handle complex scheduling, overlapping events, and dynamic time-based queries.

ACHIEVEMENTS

Academics: Achieved SOF Olympiad Gold Medals in English, Science, and General Knowledge, alongside a Silver Medal in Mathematics (6th Grade).

Sports: Winner of the Sahodaya Handball U-16 Men's tournament (2022); Won Salem district-level Chess tournaments (11th Grade).

Extracurriculars: Secured 3rd place in an intracollege group dance competition (2024).

SKILLS

Languages	Python, C, C++
Frameworks & Libraries	OpenCV, TensorFlow, PyTorch, FEniCSx, Pandas, NumPy, Scikit-Learn Plotly, Dash, Streamlit, FAISS
Tools & Platforms	VS Code, Git, GitHub, GCP, Webots, STM32, Gmsh, Blender
Core Competencies	Data Structures & Algorithms, Machine Learning, Computer Vision, Signal Processing, Cryptography/RSA, FEM Mesh Generation, Interval Trees

CERTIFICATIONS

Google IT Automation with Python *[Link]*